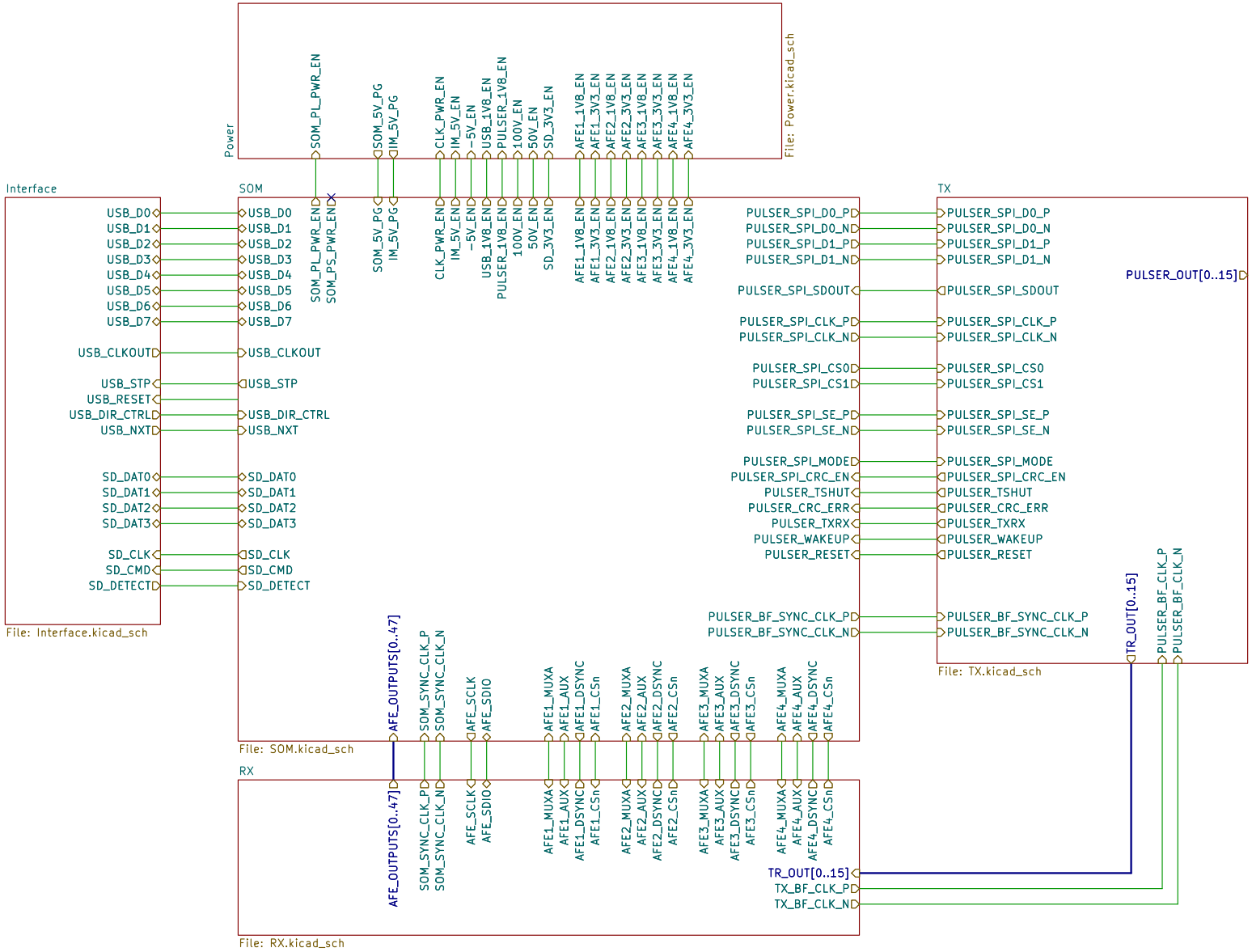
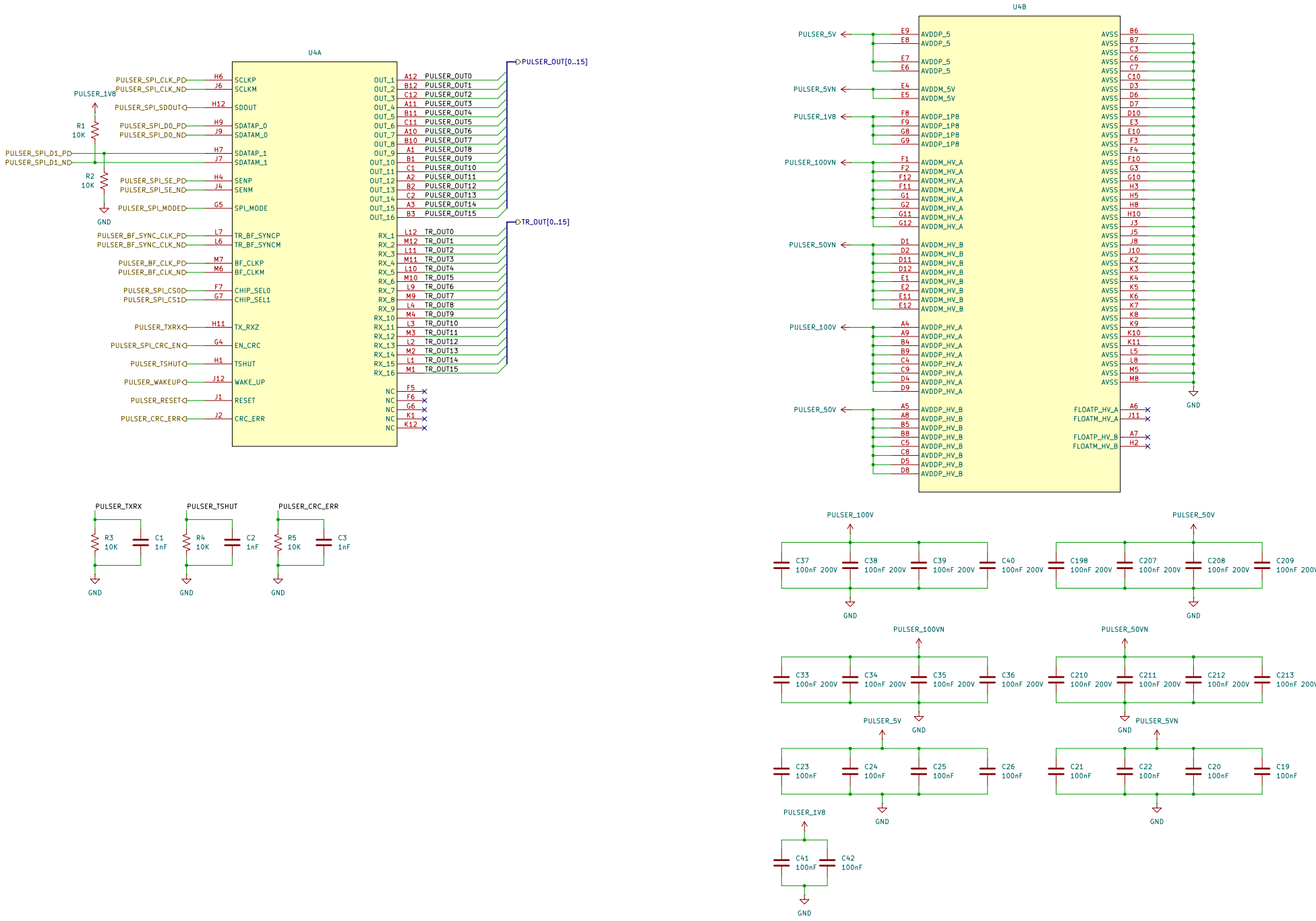
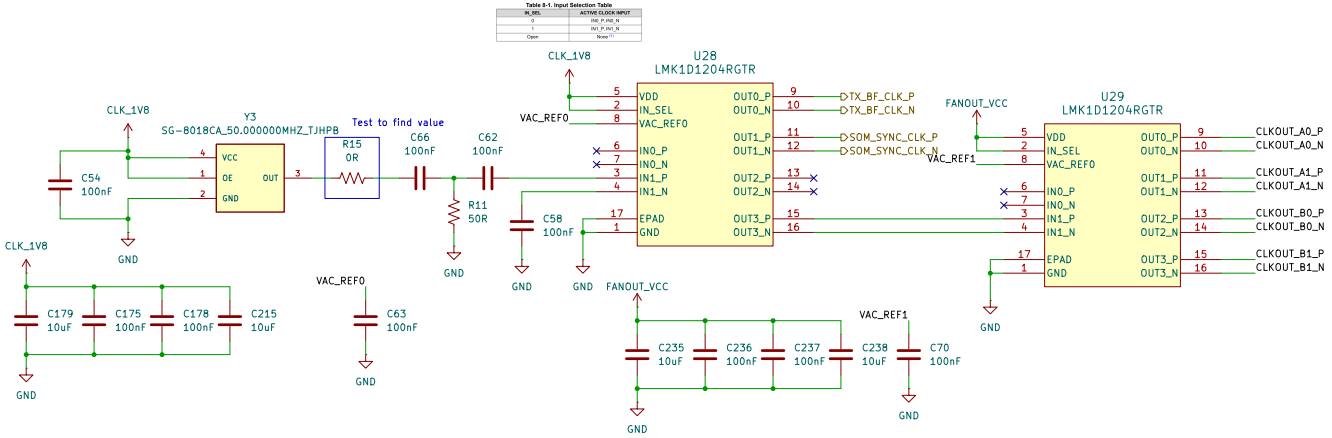
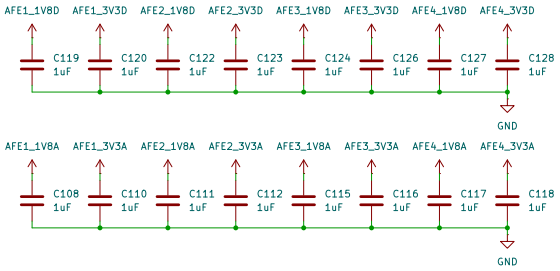
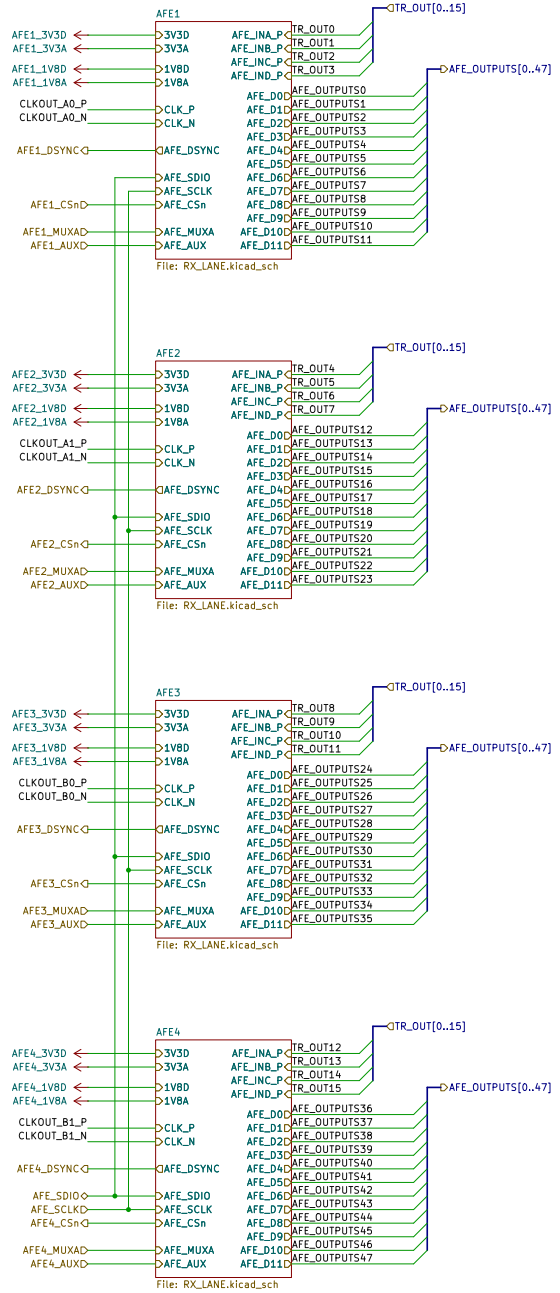


– FINALIZE CLOCKING CONNECTIONS AND TERMINATION THROUGHOUT THE CIRCUIT
– VERIFY SIGNAL CHAIN COMPATIBILITY


U4A
PULSER_SPI_CLK_PD
PULSER_SPI_CLK_ND
PULSER_1V8
PULSER_SPI_SDOUT
R1
10K
PULSER_SPI_DO_PD
PULSER_SPI_DO_ND
PULSER_SPI_D1_PD
PULSER_SPI_D1_ND
R2
10K
PULSER_SPI_SE_PD
PULSER_SPI_SE_ND
PULSER_SPI_MODE
GND
PULSER_BF_SYNC_CLK_PD
PULSER_BF_SYNC_CLK_ND
PULSER_BF_CLK_PD
PULSER_BF_CLK_ND
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_TXRX
PULSER_SPI_CRC_EN
PULSER_TSHUT
PULSER_WAKEUP
PULSER_RESET
PULSER_CRC_ERR
SCLKP
SCLKM
SDOUT
SDATAP_0
SDATAM_0
SDATAP_1
SDATAM_1
SENP
SENM
SPI_MODE
TR_BF_SYNCP
TR_BF_SYNCM
BF_CLKP
BF_CLKM
CHIP_SEL0
CHIP_SEL1
TX_RXZ
EN_CRC
TSHUT
WAKE_UP
RESET
CRC_ERR
PULSER_OUT[0..15]
PULSER_OUT0
PULSER_OUT1
PULSER_OUT2
PULSER_OUT3
PULSER_OUT4
PULSER_OUT5
PULSER_OUT6
PULSER_OUT7
PULSER_OUT8
PULSER_OUT9
PULSER_OUT10
PULSER_OUT11
PULSER_OUT12
PULSER_OUT13
PULSER_OUT14
PULSER_OUT15
TR_OUT[0..15]
TR_OUT0
TR_OUT1
TR_OUT2
TR_OUT3
TR_OUT4
TR_OUT5
TR_OUT6
TR_OUT7
TR_OUT8
TR_OUT9
TR_OUT10
TR_OUT11
TR_OUT12
TR_OUT13
TR_OUT14
TR_OUT15
NC
PULSER_TXRX
R3
10K
C1
1nF
GND
PULSER_TSHUT
R4
10K
C2
1nF
GND
PULSER_CRC_ERR
R5
10K
C3
1nF
GND
U4B
PULSER_5V
PULSER_5VN
PULSER_1V8
PULSER_100VN
PULSER_50VN
PULSER_100V
PULSER_50V
AVDDP_5
AVDDM_5V
AVDDP_1P8
AVDDM_HV_A
AVDDP_HV_A
AVDDM_HV_B
AVDDP_HV_B
AVSS
FLOATP_HV_A
FLOATM_HV_A
FLOATP_HV_B
FLOATM_HV_B
GND
PULSER_100V
C37
100nF 200V
C38
100nF 200V
C39
100nF 200V
C40
100nF 200V
GND
PULSER_50V
C198
100nF 200V
C207
100nF 200V
C208
100nF 200V
C209
100nF 200V
GND
PULSER_100VN
C33
100nF 200V
C34
100nF 200V
C35
100nF 200V
C36
100nF 200V
GND
PULSER_50VN
C210
100nF 200V
C211
100nF 200V
C212
100nF 200V
C213
100nF 200V
GND
PULSER_5V
C23
100nF
C24
100nF
C25
100nF
C26
100nF
GND
PULSER_5VN
C21
100nF
C22
100nF
C20
100nF
C19
100nF
GND
PULSER_1V8
C41
100nF
C42
100nF
GND

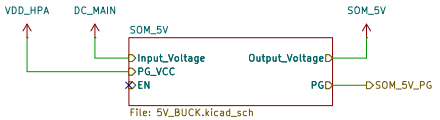






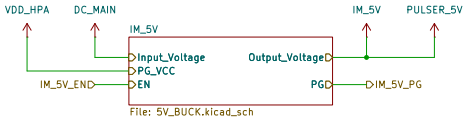
AFE IO are 3V3

Main SOM 5V Supply



Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.2V – 1.98V	100uA – 500uA	Coin cell battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44

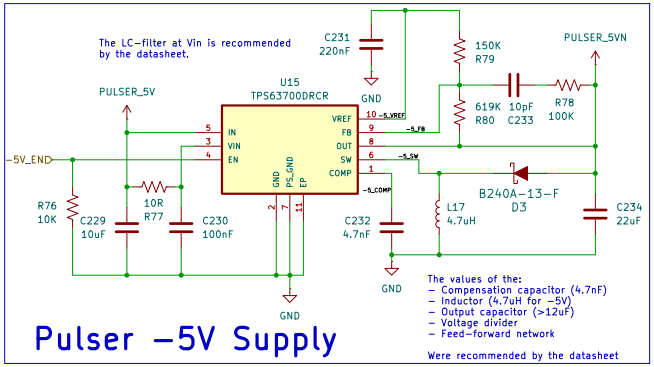
Pulser + Intermediate 5V Supply

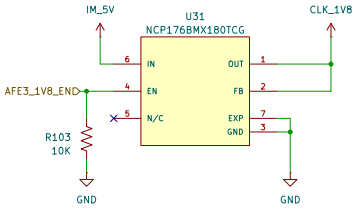


$I_{MODIO18}$	Full channel mode	130	mA
$I_{MODIO33}$	Full channel mode	130	mA
$I_{DVDIO18}$		22	mA
$I_{DVDIO33}$		2	mA

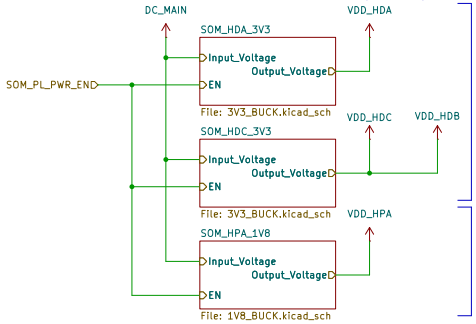
^ x4 = 1.2A

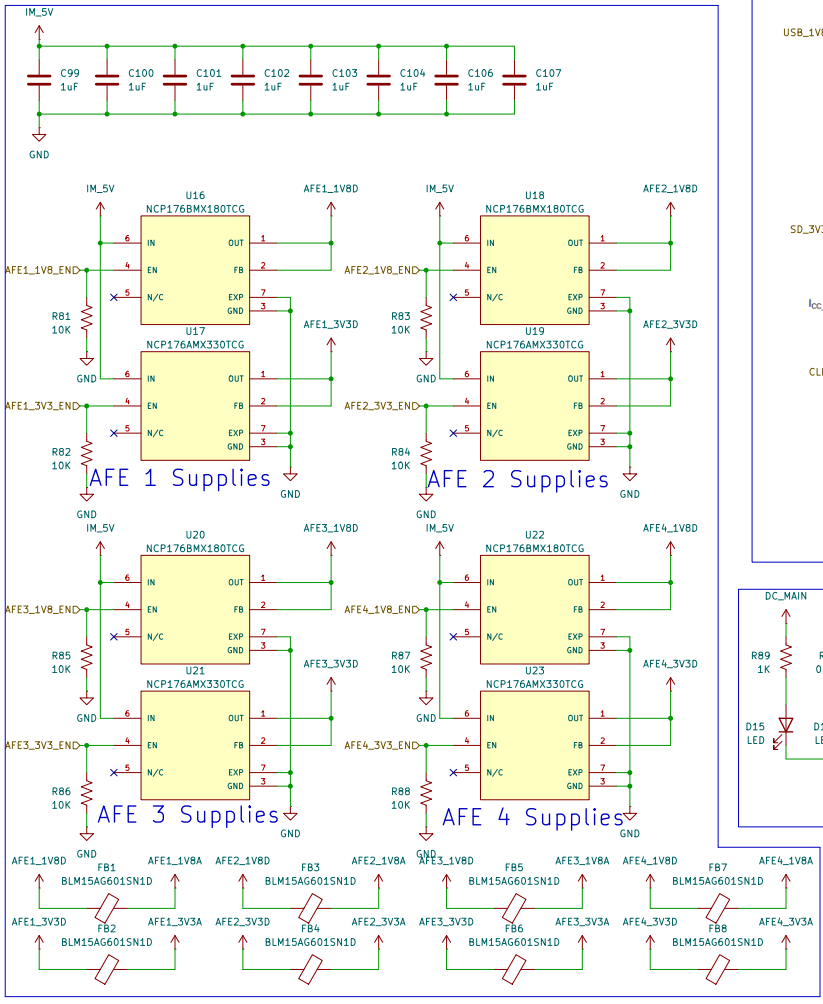
Pulser −5V Supply



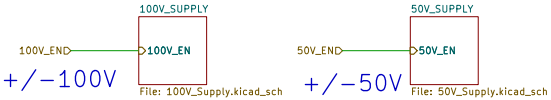


Processing Section (PL) Supplies

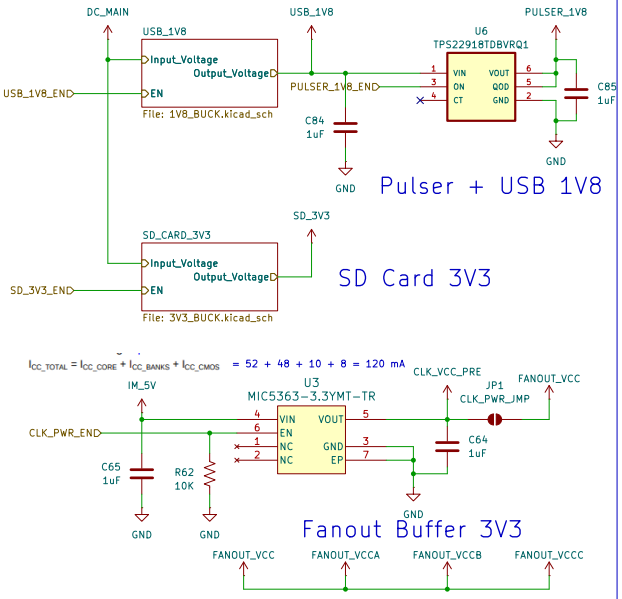


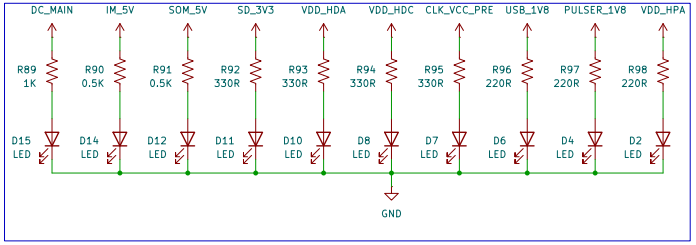


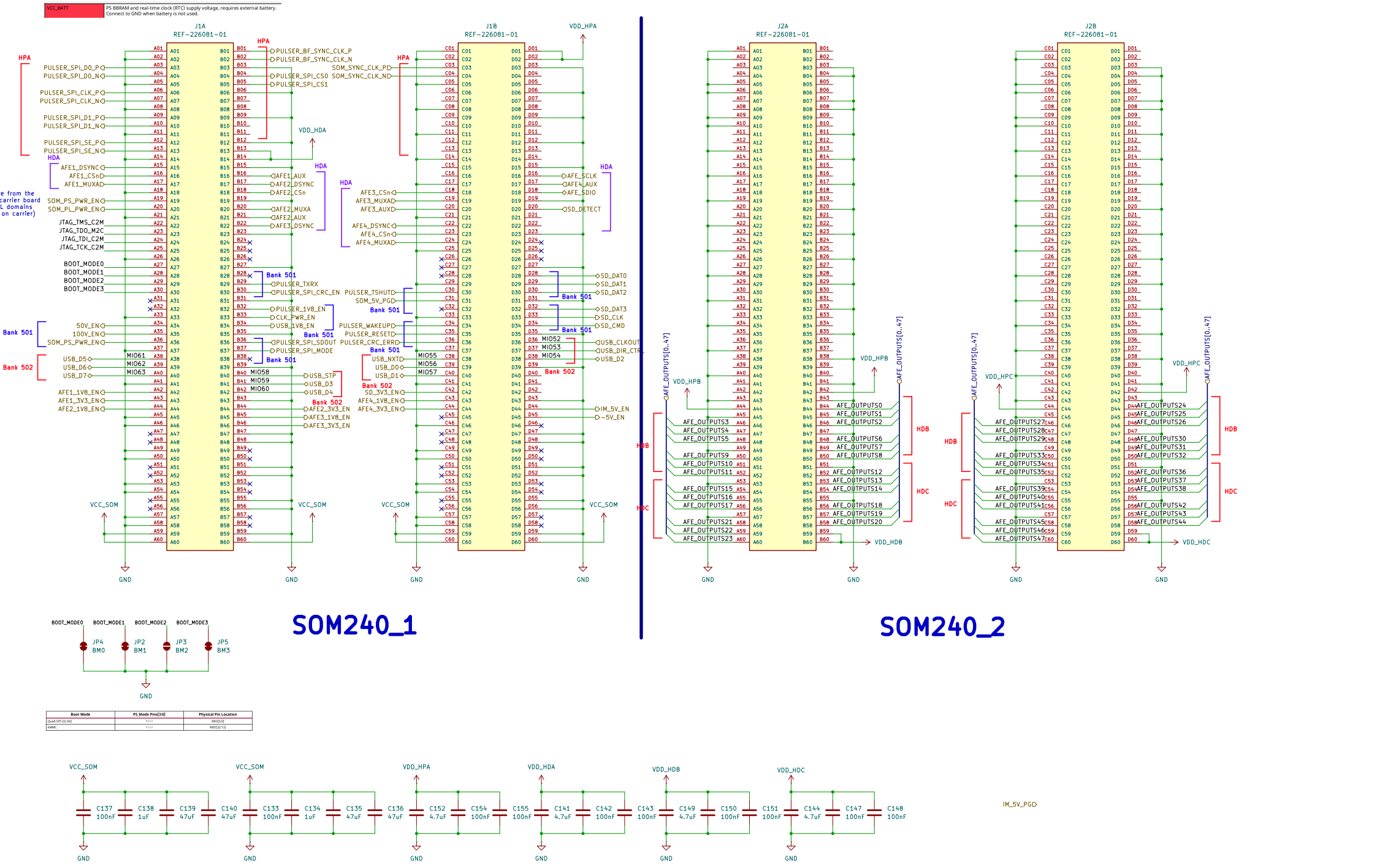
HV Supplies (+/− 50/100V)

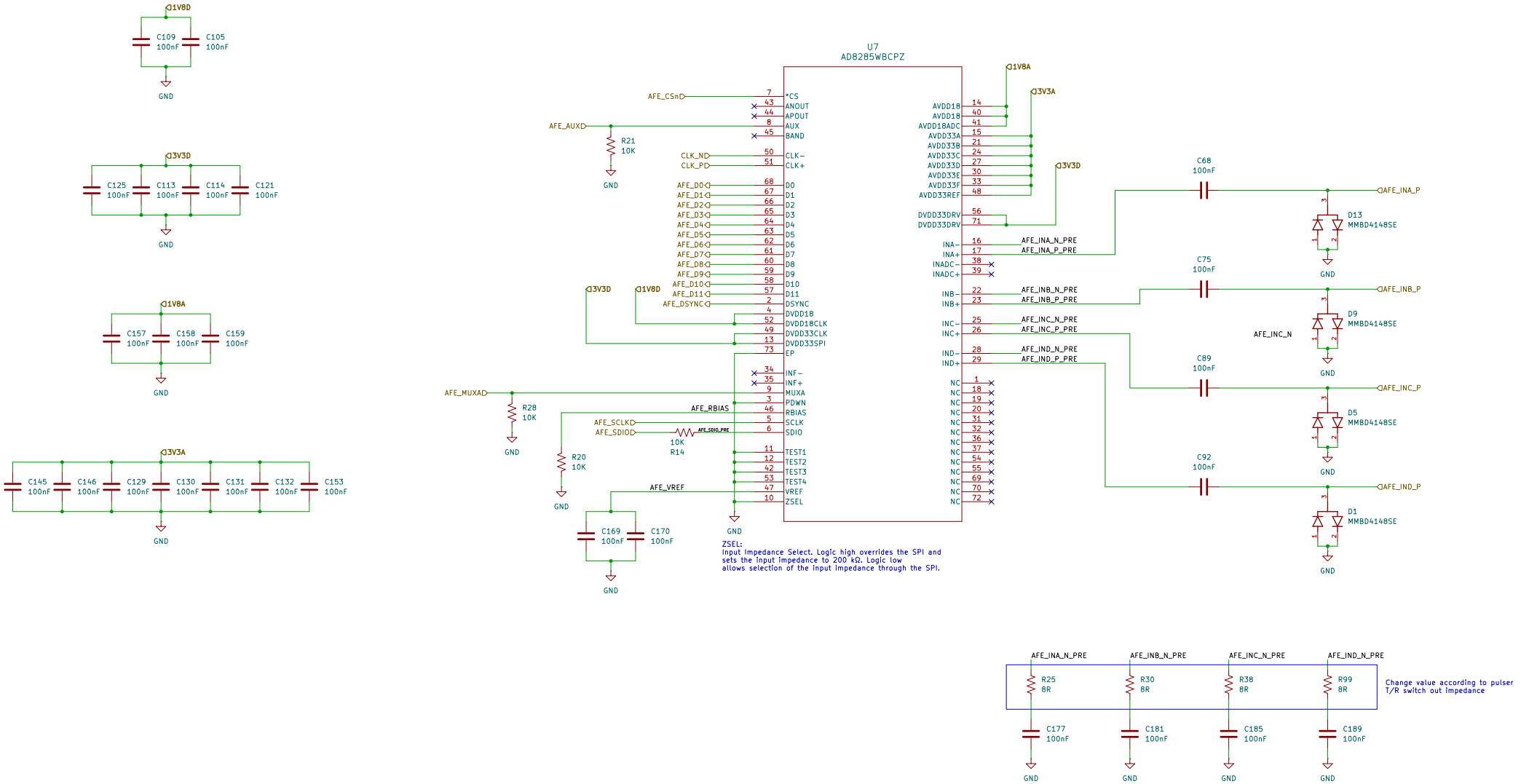


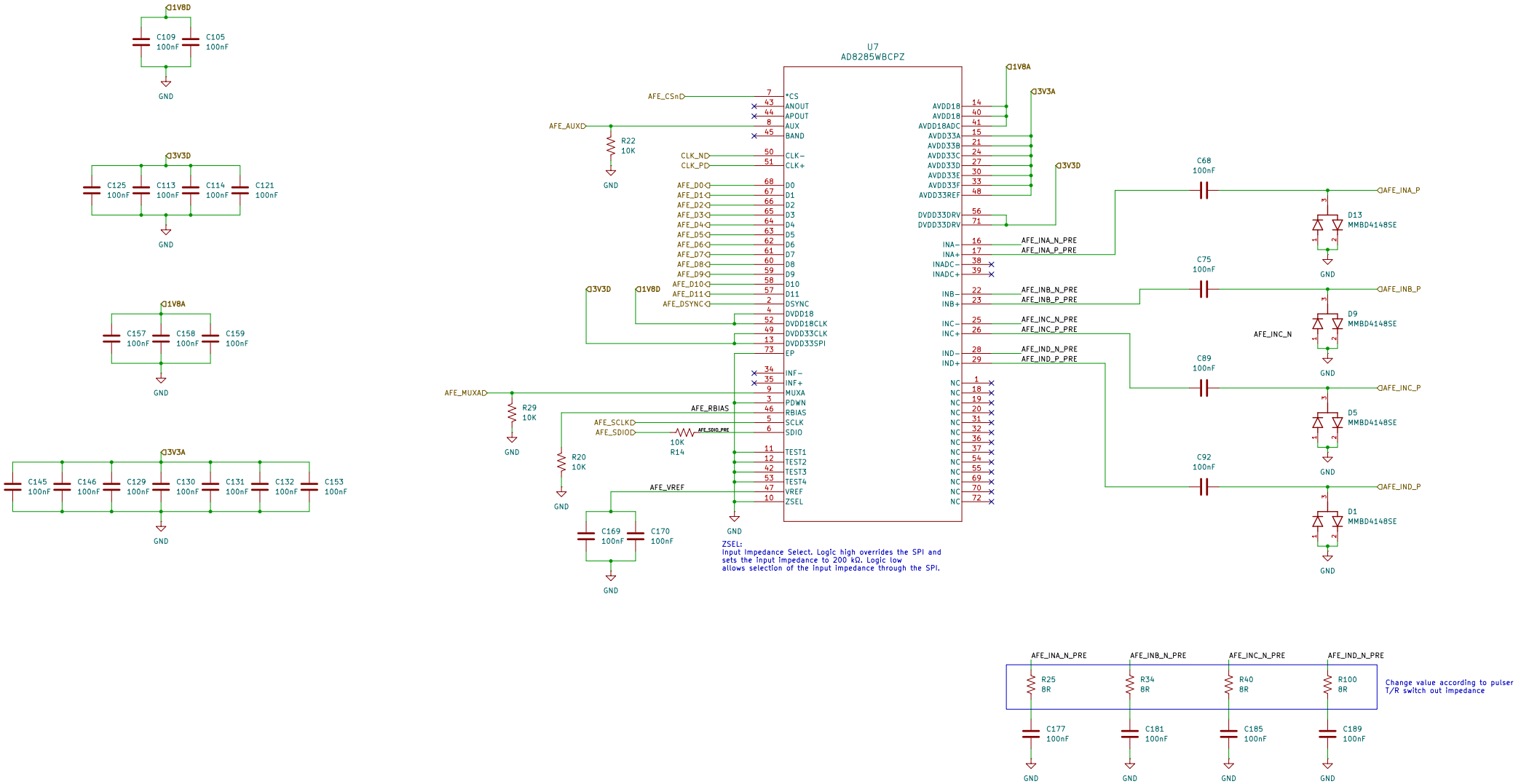
Miscellaneous Supplies

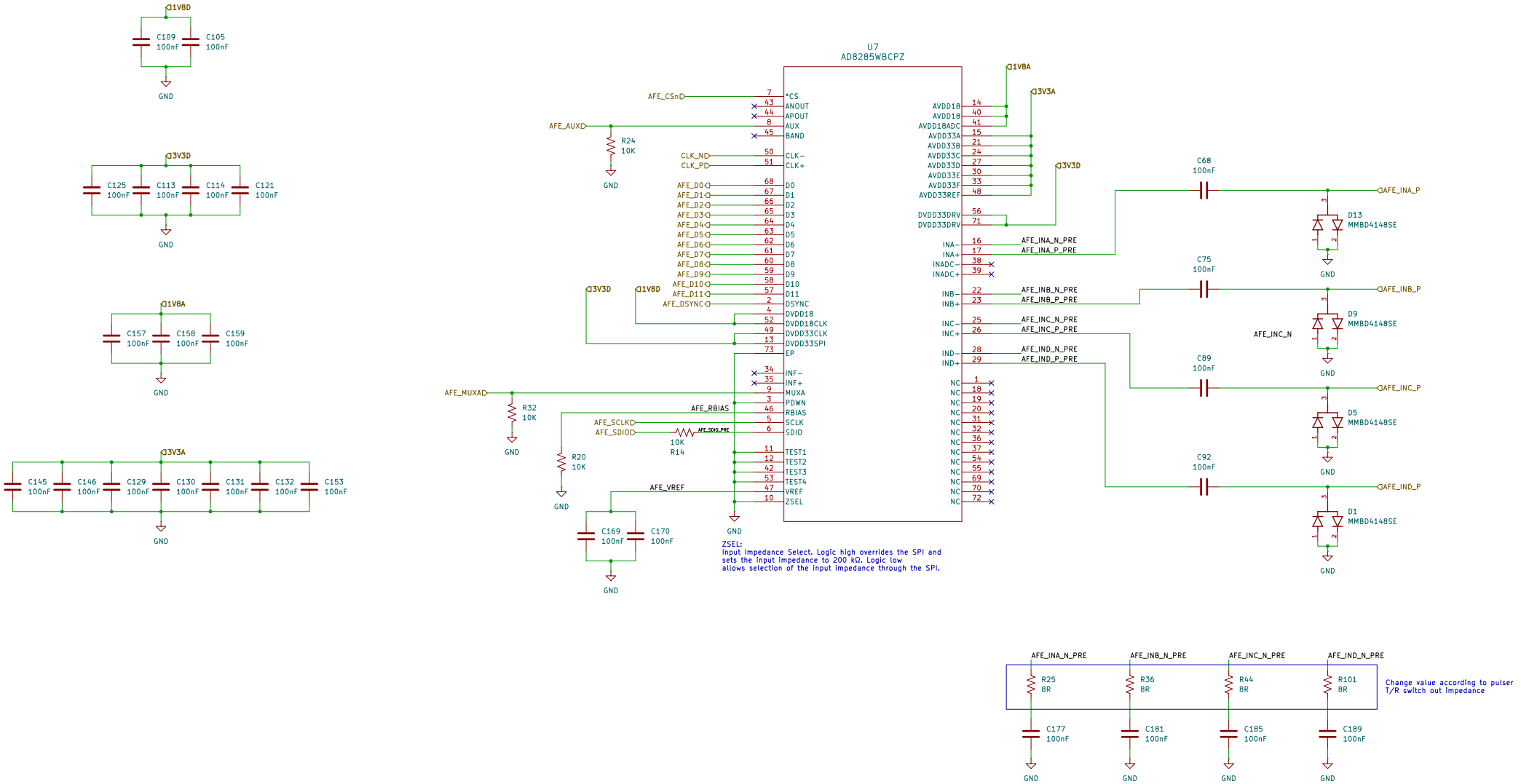


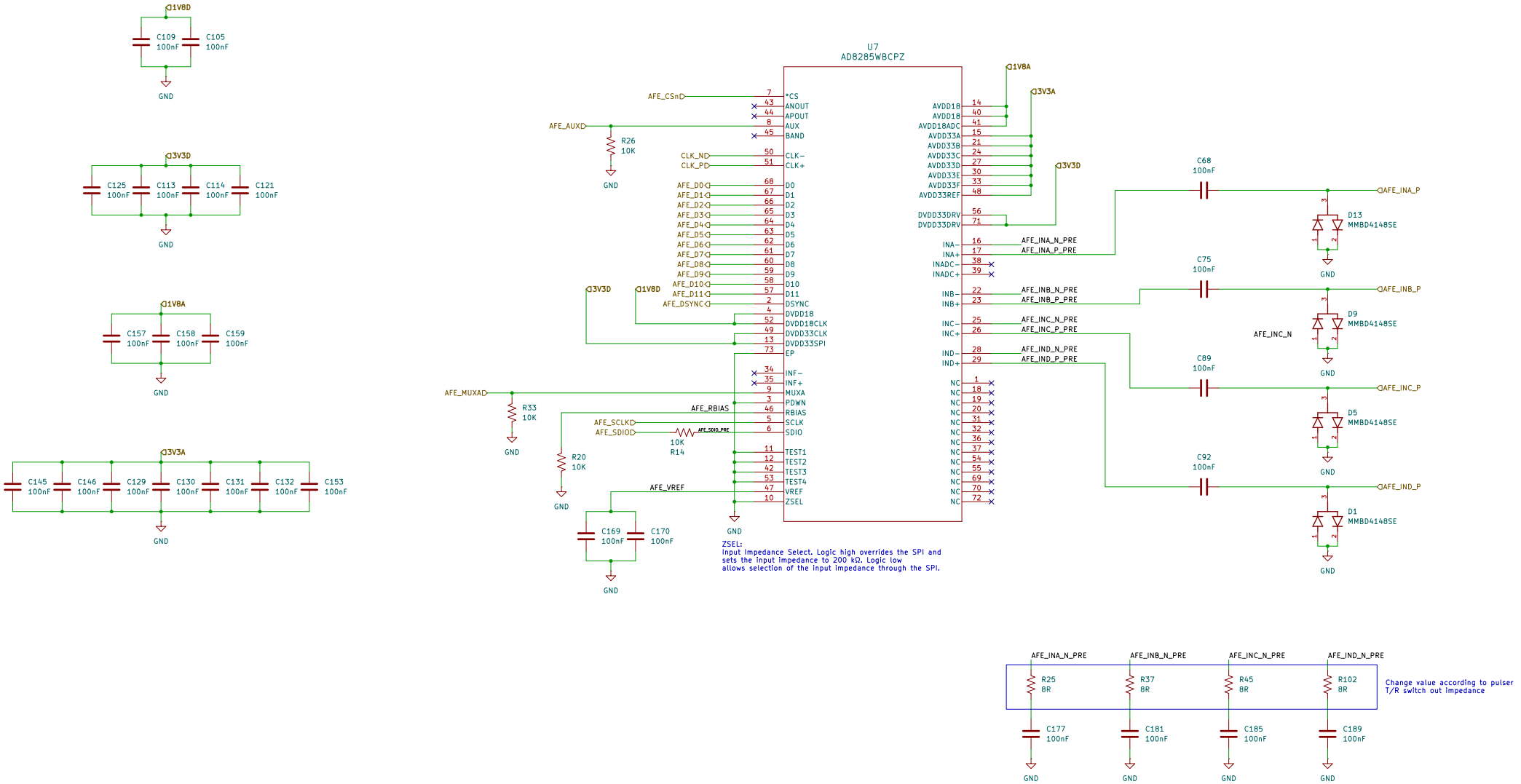



VCC_BATT
PS BBRAM and real-time clock (RTC) supply voltage, requires external battery.
Connect to GND when battery is not used.
J1A
REF-226081-01
J1B
REF-226081-01
J2A
REF-226081-01
J2B
REF-226081-01
SOM240_1
SOM240_2
BOOT_MODE0
BOOT_MODE1
BOOT_MODE2
BOOT_MODE3
JP4 BM0
JP2 BM1
JP3 BM2
JP5 BM3
Boot Mode
PS_Mode Pins[3:0]
Physical Pin Location
Quad-SPI (32 bit)
eMMC
VCC_SOM
VDD_HPA
VDD_HDA
VDD_HPB
VDD_HDB
VDD_HPC
VDD_HDC
GND
IM_5V_PGD


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R21 10K
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
AFE_MUXA
R28 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
ZSEL:
Input impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
R25 8R
R30 8R
R38 8R
R99 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF

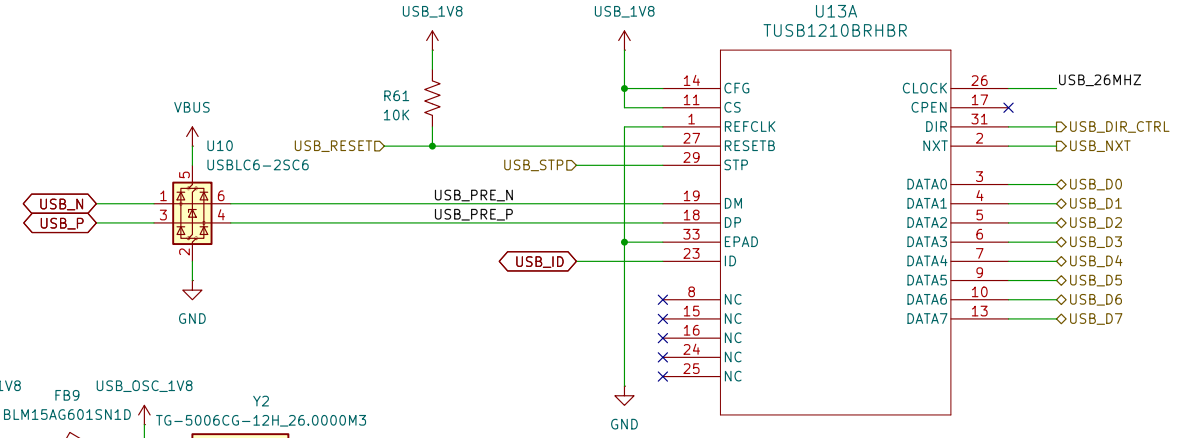

1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R22 10K
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
AFE_MUXA
R29 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
C68 100nF
C75 100nF
C89 100nF
C92 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R34 8R
R40 8R
R100 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF

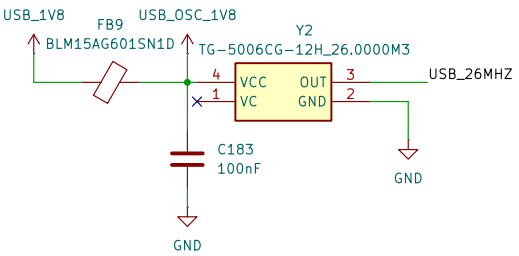

1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R24 10K
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
AFE_MUXA
R32 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
C68 100nF
C75 100nF
C89 100nF
C92 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R36 8R
R44 8R
R101 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF

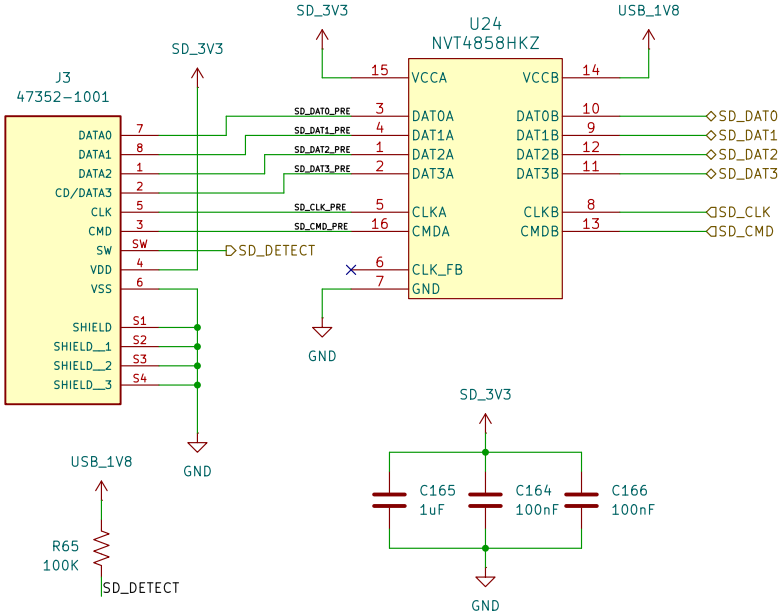

1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R26 10K
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
AFE_MUXA
R33 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
R25 8R
R37 8R
R45 8R
R102 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF

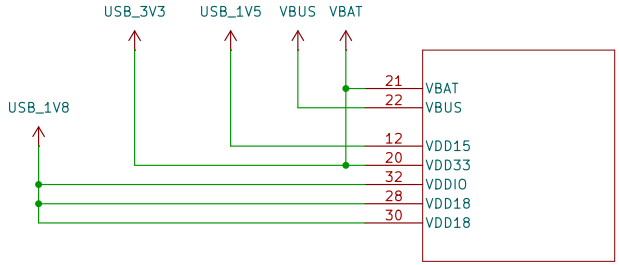
CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.

Check AN1310 for layout guidelines

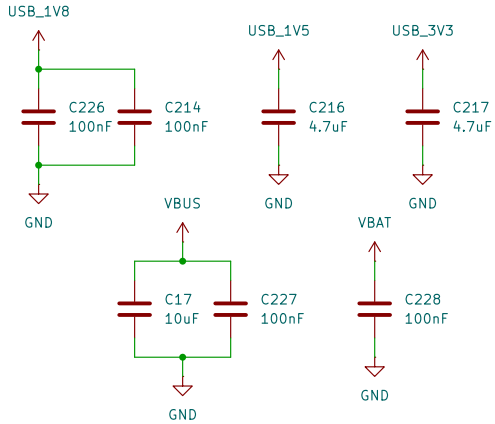


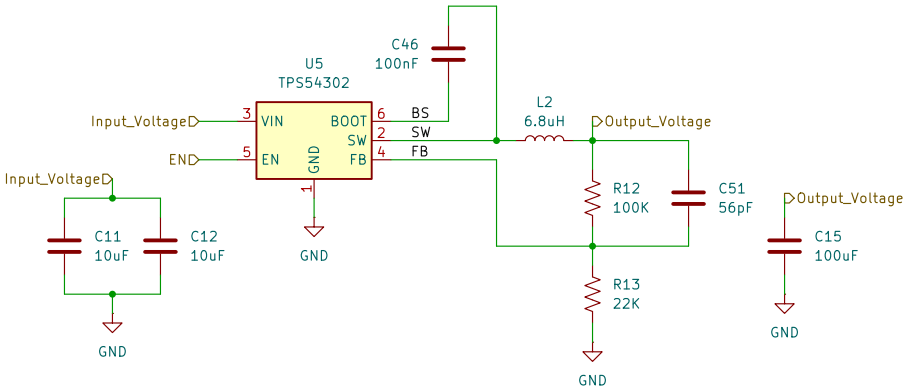




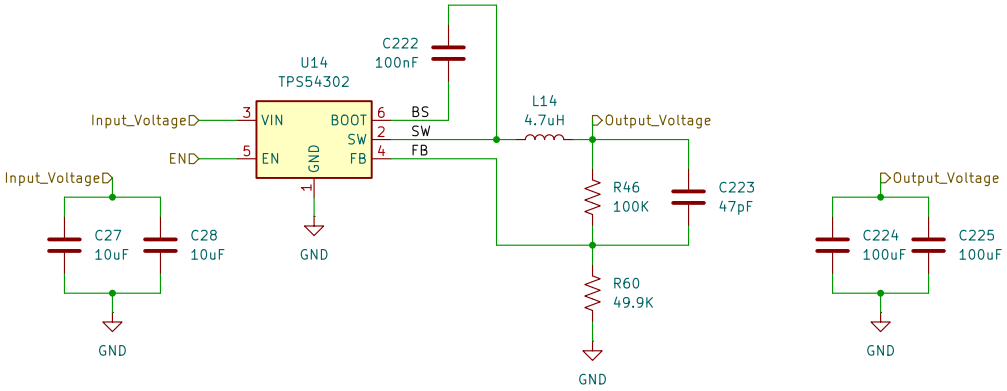


3V3 & 1V5 are the outputs of internal LDOs





WEBENCH DESIGNER LINK (3.3V)



WEBENCH DESIGNER LINK (1.8V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

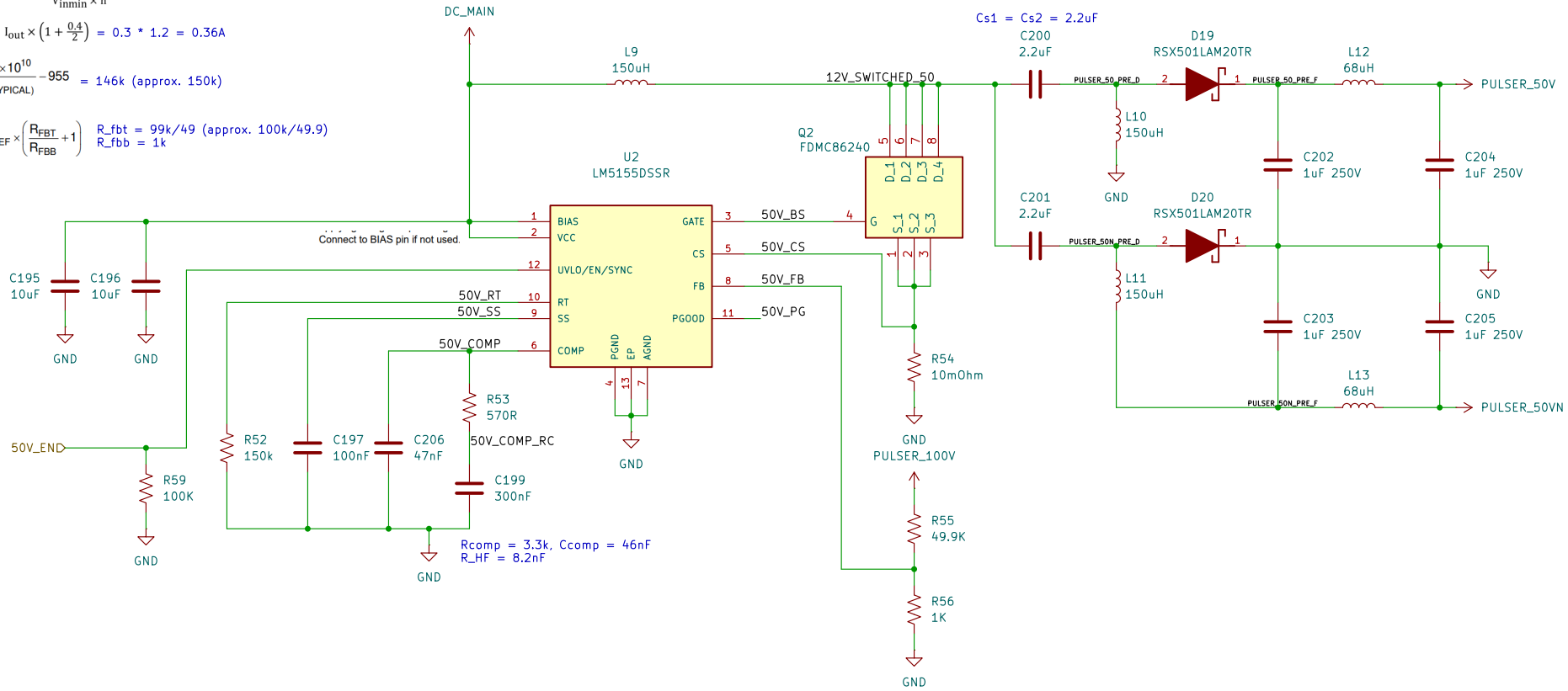
n = 0.88

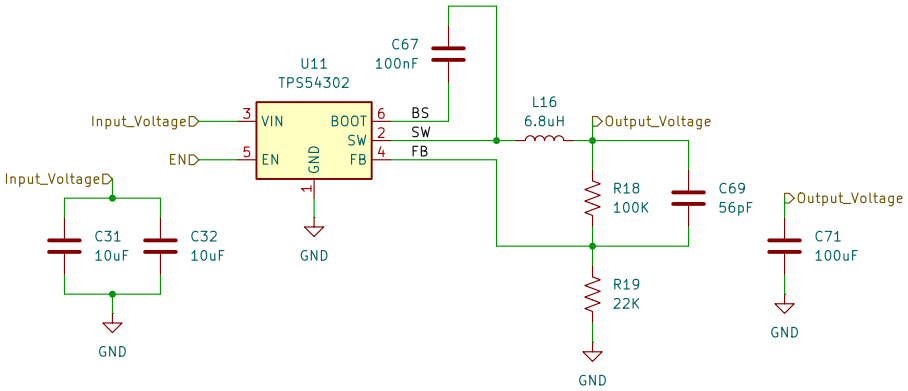
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k





WEBENCH DESIGNER LINK (3.3V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

n = 0.88

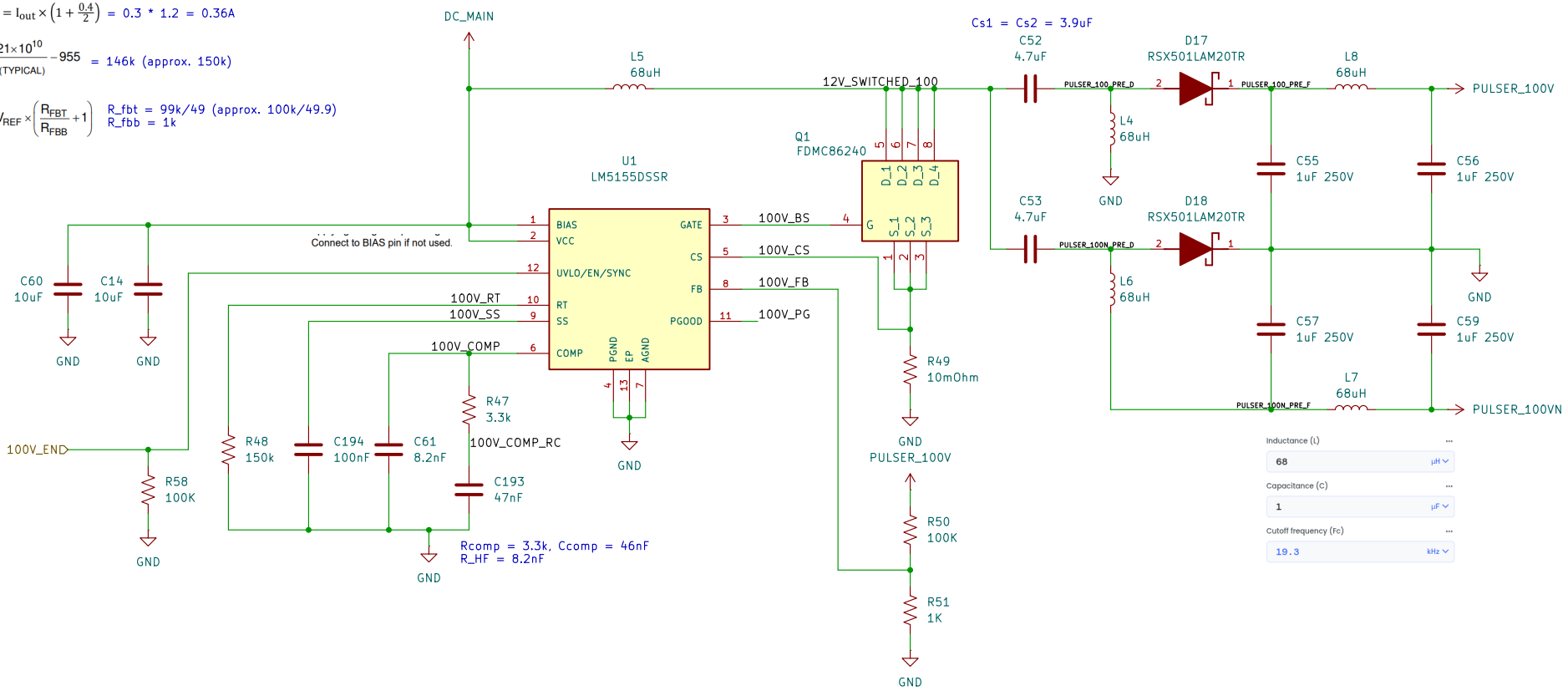
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

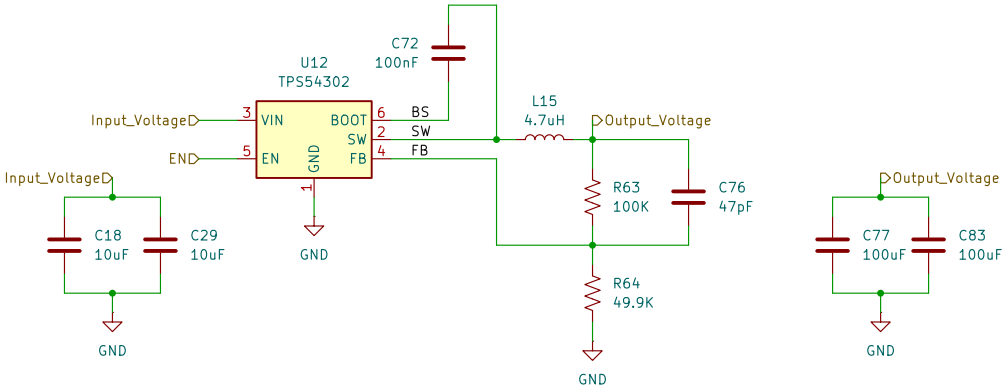
$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

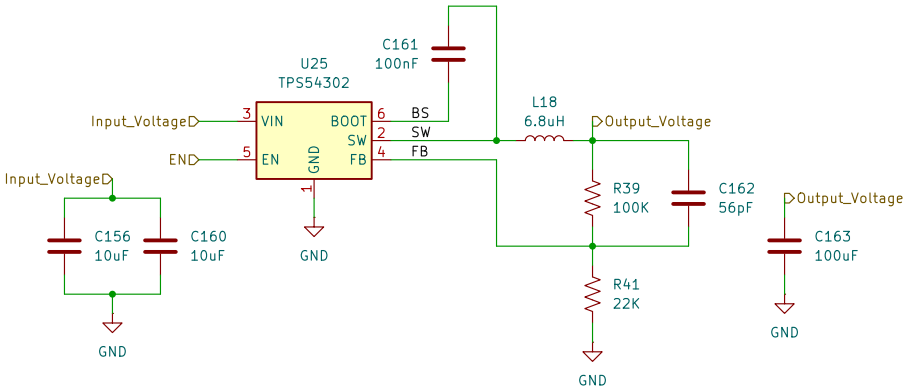
$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)

R_fbb = 1k

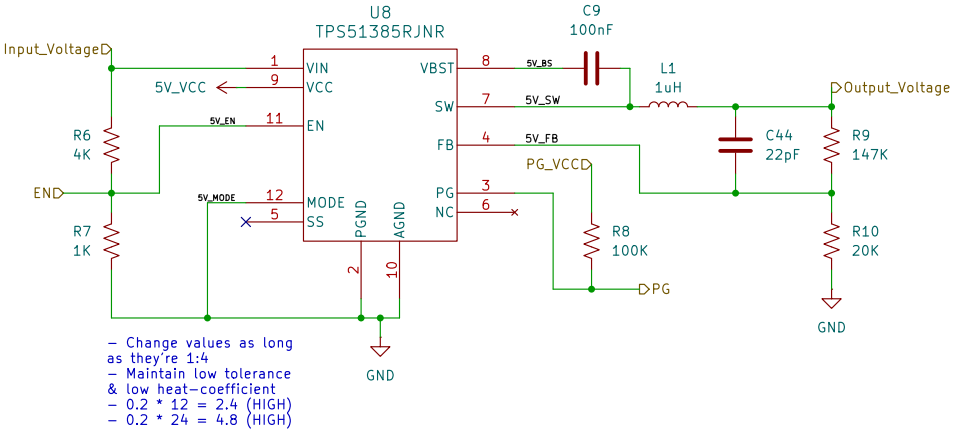


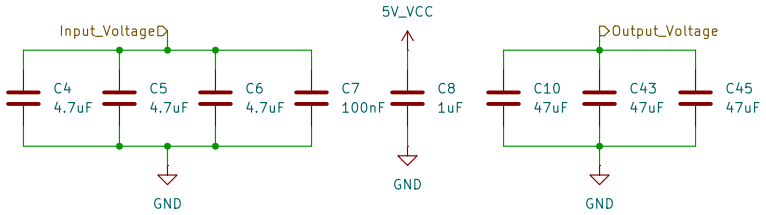


WEBENCH DESIGNER LINK (1.8V)

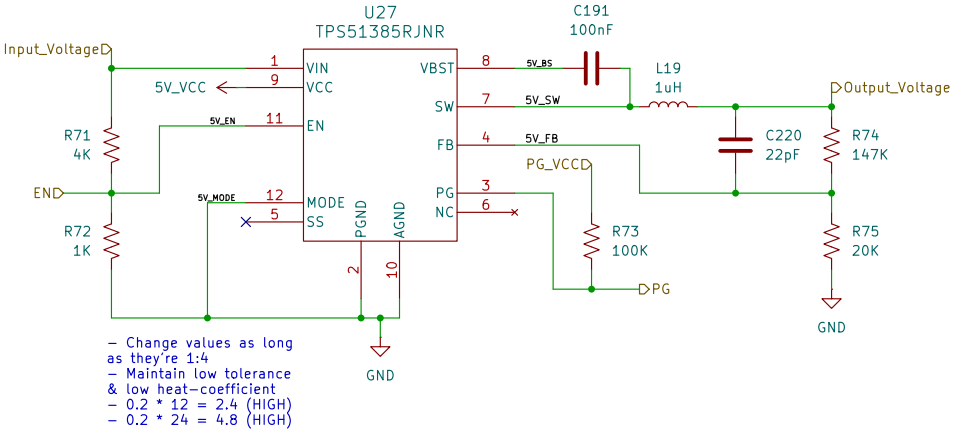


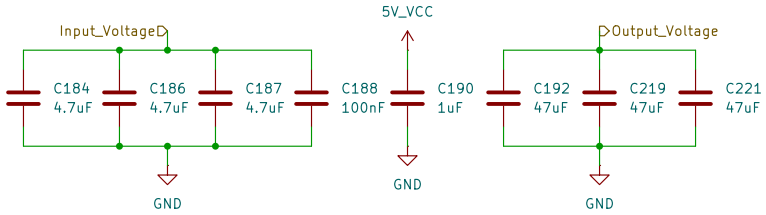
WEBENCH DESIGNER LINK (3.3V)





WeBench Designer Link





WeBench Designer Link